Common Conditions Impacting Older Adults

- Arthritis: When your patient has arthritis, provide extra padding by using a towel or pillow to cushion their arm to help increase their comfort level. Avoid moving your patient quickly or abruptly to reduce discomfort. Ask your patient if they are comfortable placing themselves in the proper position for procedures. If they are unable to achieve a certain position comfortably, work with them to find a position that is comfortable for them, always considering both the patient's safety and your own.
- Hearing impairment: When a patient has some level of hearing loss, ensure they have heard your instructions by asking them to repeat back the instructions. When in doubt, it is better to ask for a translator or to write down the instructions. Raise your voice in conversation only when you know that your patient has hearing loss from their patient record or when they mention that they can't hear what you said.
- Visual impairment: Try to keep the area around your worksite clear of clutter to reduce tripping hazards. Do not get in the way of a cane or working dog; your patient uses these as visual aids, and interfering with them can negatively impact their navigation. Use verbal cues to help them maneuver through complex areas.
- Movement impairment: If your patient uses a wheelchair, it may not be necessary to move them to a blood collection chair. Once you are assured that the brakes on the wheelchair are set, you can perform a venipuncture without taking the risk of moving your patient. If your patient uses a cane, avoid getting in the way of its use. Make sure that it is placed close to the patient if you must move it for the venipuncture. Do not grab, lean on, or use these devices. Treat a cane or wheelchair as part of your patient.
- Memory impairment: Demonstrate patience and be willing to repeat answers to your patient's questions. Sometimes it helps to speak to your patient to confirm that they are aware of what you are doing throughout the procedure. When in doubt of your patient's ability to provide consent, work with their nurse to help ensure that your patient understands the procedure and what actions you need them to perform.

Do Not Make Assumptions

Do not assume that an older adult experiences any limitations. Speak in your usual tone until your patient expresses that they cannot hear you. Starting a conversation with an older adult patient by yelling at them, especially when they are able to hear, can cause aggravation that may further complicate communication with them. In addition, do not assist them unless asked to help foster their dignity and independence.

To help place an infant in position for blood collection, place them on their side to help reduce the risk of choking. Keeping an infant warm and secure, by swaddling them, for example, helps increase blood flow. It also helps your patient stay still enough for the blood collection to be performed safely. If the parents or guardians use a pacifier or are nursing, either can be used during the collection to help relax the infant and keep them still. Communication with children is helpful in gaining their trust and ensuring their adherence to your instructions.

Collection From Children

When performing venipunctures on children, it is important to monitor the amount of blood being collected. Avoid collecting more than 10% of an infant's blood volume in a short period of time. Collect no more than 1% to 5% of a child's or infant's blood volume over a 24-hour period of time to help prevent health issues such as iatrogenic anemia or cardiac arrest.

To help determine how much blood can safely be collected from an infant or child, you must first calculate their blood volume.

- Convert the infant's weight from pounds to kilograms.
- Divide pounds by 2.2.
  For example, 6.2 lb ÷ 2.2 = 2.82 kg.
- Multiply this sum by 100 to calculate blood volume in milliliters. For example, 2.82 × 100 = 282 mL.
- Convert blood volume from milliliters to liters. For example, 282 mL ÷ 1,000 = 0.28 L.

Do Not Make Assumptions

When performing a venipuncture on either an older adult, infant, or child, do not assume that you need to use a butterfly or syringe just because of a person's age. Evaluating their veins, not their age, to help determine what type of needle, tube, and procedure will help you successfully collect a sample.

4.1 CHALLENGE

You are preparing to collect blood from three patients. For each patient, assess the potential blood collection sites. Keep in mind that each patient has their own complications which can affect where you should perform the collection.

1. Complication: Burn covering most of the left arm and hand with some scarring
   A. Right hand: visible and palpable
   B. Left hand: visible and palpable
   C. Right cephalic: not visible or palpable
   D. Left cephalic: visible and palpable
   E. Right median cubital: not visible nor palpable
   F. Left median cubital: visible and palpable
   G. Right basilic: visible but not palpable
   H. Left basilic: not visible nor palpable
2. Complication: CVAD located in the upper right forearm with some tortuous veins present on the lower left forearm and hand
   A. Right hand: not visible nor palpable
   B. Left hand: visible and palpable
   C. Right cephalic: visible and palpable
   D. Left cephalic: visible and palpable
   E. Right median cubital: visible but not palpable
   F. Left median cubital: visible and palpable
   G. Right basilic: visible and palpable
   H. Left basilic: palpable but not visible

4.1 CHALLENGE KEY

1. A. Since the patient has significant burns covering most of the left arm and hand, this side should be avoided. The right cephalic and median cubital are neither visible nor palpable, and the right basilic is only visible, but not palpable. Therefore, the best option for this venipuncture is the dorsal side of the right hand.
2. F. Since the patient has a CVAD on the right side, this side should be avoided. While the patient does have some tortuous veins in the left forearm, they are located distally and will not interfere with use of the antecubital space. Since both the median cubital and cephalic are visible and palpable on the left, the left median cubital should be used since this vein has the lowest chance of pain and complications.

Conclusion

Proper preparation of yourself and your supplies and knowing the proper sites for venipuncture sets the stage for a successful interaction with your patient. Confirming that your supplies are in proper operating condition helps ensure accurate test results. Gathering all necessary supplies before beginning the blood collection process helps minimize extra work and reassures patients of your professionalism. Evaluate your patient for any conditions that may require special attention to help ensure a safe and accurate blood collection. Take all necessary measures to help confirm that your patient is safe throughout every procedure. Review requisitions to help familiarize yourself with the often-ordered tests and their matching tubes. Practice tying and releasing a tourniquet to build confidence and ensure you can do so comfortably without hurting or causing discomfort to your patient. Choose the best vein every time to help reduce the risk of injury to your patient and ensure a successful venipuncture.

4.2 CHALLENGE

1. Place these sites in the best order for venipuncture.
   - A. Dorsal part of the hand
   - B. Median cubital
   - C. Foot or ankle
   - D. Basilic
   - E. Cephalic
2. Knowing how to dispose of used supplies properly is important for all health care professionals. You have the choice of three disposal methods: a sharps container, a biohazard bin, and a trash can. Drag and drop the used materials to the correct disposal container.
   1. Sharps container
   2. Trash can
   3. Biohazard bin

   - A. Evacuated tube needle
   - B. Disposable gown
   - C. Glucose testing stripe
   - D. Alcohol cleaning pad
   - E. Butterfly needle
   - F. Evacuated tube
   - G. Alcohol cleaning pad package
   - H. Glucose testing strip container
   - I. Cotton ball
   - J. Syringe
   - K. Disposable gloves
   - L. Plastic pipette
   - M. Evacuated tube
   - N. Plastic aliquot tube
   - O. Disposable tourniquet

4.2 CHALLENGE KEY

1. B, E, A, D, C. The median cubital is the first choice as it is typically in the middle, well anchored, and easier to access with a needle. The basilic is the last choice in the antecubital region due to the chance of accidentally puncturing an artery or nerve.
2. Sharps container: A, E, F, J, M.
   Biohazard bin: B, C, I, K, L, N.
   Trash can: D, G, H, O.